

Frontier Methods, New Data, And Evolving Measurement

Gender Study — Week 9

Teaching deck draft

Central question

- How are new data, designs, and measurement choices changing gender-and-labor research?
- Where are the main empirical shortfalls in the existing literature?
- How do we turn those shortfalls into research designs?

labor object → data family → observed margin → identification → welfare interpretation.

Why methods matter for gender and labor

- Gender gaps are not one outcome: employment, hours, applications, wages, promotion, authority, safety, climate, identity, and welfare all matter.
- Better data reveal new margins but do not automatically identify mechanisms.
- The same substantive question may require different data: admin panels, matched worker-firm records, audits, postings, platforms, surveys, or time use.
- Week 9 is a research-career bridge: find the observed margin, then name what remains missing.

Bridge from Weeks 2–8

Week	Substantive object	Week 9 methods question
2	care, fertility, household allocation	Which event, time-use, or policy variation isolates the care margin?
3	education, aspirations, sorting	Do we observe expectations, applications, or only realized choices?
4	firms, promotion, pay-setting	Is the gap worker sorting or firm treatment?
5	norms and bargaining	Is the norm measured or inferred from outcomes?
6	policy incidence	Which workers, firms, and households bear the incidence?
7	safety and hidden harms	Do we observe exposure, reporting, or labor response?
8	comparative mechanisms	What transports across institutional settings?

Descriptive gap vs causal identification

Descriptive gap

$$\Delta^Y = E[Y_i \mid G_i = 1] - E[Y_i \mid G_i = 0]$$

- Locates where gender differences appear.
- Requires clarity about outcome, population, and gender measure.

Causal object

$$\tau = E[Y_i(1) - Y_i(0) \mid i \in \mathcal{P}]$$

- Requires a counterfactual.
- Requires identifying variation and a target population.

Data-family taxonomy

Data family	Margin seen well	Persistent blind spot
Matched worker–firm	firm premiums, moves, within/between gaps	nonapplications, hidden harms
Administrative panels	lifecycle dynamics, policy timing	care burden, climate, beliefs
Audits/field experiments	screening, callbacks, evaluation	long-run careers, equilibrium response
Surveys/expectations	beliefs, norms, anticipated discrimination	persistence, response selection
Time-use data	unpaid work, care, scheduling	firm treatment if unlinked
Postings/text	employer demand, language, stated constraints	realized treatment after hire
Platform data	high-frequency supply, tasks, pay	outside options, nonplatform careers
Identity/climate	lived treatment, belonging, exposure	small cells, privacy, comparability

Setting-to-methods map

Empirical setting	Observed margin	Practical tools
Childbirth or family shock	earnings, hours, employment event time	event studies, dynamic treatment effects
Matched worker–firm panel	worker moves, wages, firm identifiers	worker and firm fixed effects, mover designs
Resume or hiring audit	callback, interview, offer	randomized signals, heterogeneity
Job-board reform	applications, callbacks, match composition	difference-in-differences, ad-level event studies
Time-use diary	care time, market hours	time-budget decompositions
Platform panel	hours, timing, tasks, earnings	within-worker decompositions
Climate or harassment linkage	complaints, exits, wages, turnover	linked survey-admin, hazards, selection diagnosis

Matched employer–employee methods

- Core question: how much of the gender wage gap reflects firm sorting versus firm treatment?
- Observed margin: worker-firm matches, wages, mobility, tenure, sometimes occupation or establishment.
- Identifying variation: movers across firms and changes in pay around firm transitions.
- Tools: AKM-style worker and firm fixed effects, mover designs, within-firm and between-firm decompositions.

$$\bar{w}_m - \bar{w}_f = \sum_j (s_{mj} - s_{fj}) \bar{w}_j + \sum_j s_{fj} (\bar{w}_{mj} - \bar{w}_{fj}).$$

Administrative event-study methods

- Core question: how do earnings, hours, employment, and firm outcomes evolve around childbirth, leave, policy, or job transitions?
- Observed margin: event time in administrative panels.
- Identifying variation: timing around an event or reform.
- Tools: event studies, stacked event studies, dynamic treatment effects, outcome decompositions.
- Threats: anticipation, endogenous timing, selected event populations, and hidden care or climate margins.

Audits and field experiments

- Core question: does the screening or evaluation stage treat otherwise comparable applicants differently?
- Observed margin: callback, interview, offer, evaluation score, application response.
- Identifying variation: randomized names, gender signals, identity signals, information, or recruiting messages.
- Strength: clean treatment variation at a specific stage.
- Limit: may not observe later retention, promotion, workplace treatment, or equilibrium adjustment.

Applications, postings, and text

- Application data reveal the pre-match margin: who applies to which jobs before accepted jobs are observed.
- Posting data reveal employer demand: stated restrictions, flexibility, pay, tasks, seniority, culture signals, and language.
- Text-as-data tools include dictionaries, supervised classifiers, validated language measures, and ad-level fixed effects.
- Identifying variation can come from platform rule changes, posting bans, randomized ad text, or quasi-experimental ad exposure.

Platform data and hidden margins

- Platform records observe high-frequency behavior: hours, timing, location, task choice, speed, ratings, and earnings.
- Within-worker decompositions can separate hours, experience, task selection, and scheduling.
- The design often sees behavior conditional on platform participation.
- Hidden margins remain: outside options, safety, harassment, care constraints, algorithmic exposure, and nonplatform career effects.

Incidence

- Who bears the effect?
- Workers, firms, coworkers, applicants, households, consumers.
- Example margins: wages, hiring, compliance, sorting, hours.

Hidden harms

- What does the direct outcome miss?
- Fear, harassment, retaliation, concealment, mobility avoidance, dignity.
- A wage effect is not a complete welfare object.

Measurement of gender and identity

Measurement object	Why it matters	Main challenge
Legal sex/admin category	linkage and long panels	may not match identity or treatment
Self-identified gender	lived identity and experience	small cells, privacy, nonresponse
Perceived gender	hiring, promotion, harassment mechanism	rarely observed directly
Workplace climate	retention, authority, welfare	selective and imperfect measures
Intersectional categories	realistic heterogeneity	precision and disclosure risk
Cross-time categories	trend interpretation	forms and categories change

Legal categories vs lived treatment

- Administrative data may record legal sex because that is what the form collects.
- Employers and coworkers may respond to perceived gender, pregnancy, parenthood, name, presentation, sexuality, race-gender stereotypes, or disclosure.
- The data category and the mechanism may differ.
- A strong paper states which object is measured and which object is theoretically relevant.

Empirical challenges and shortfalls

- Selection into observed employment, firms, applications, surveys, and reports.
- Sorting versus treatment in occupations, firms, teams, policies, and platforms.
- Anticipation before childbirth, leave, promotion, reporting, and exits.
- Underreporting and hidden harms in safety, climate, and harassment research.
- Firm adaptation to transparency, quotas, compliance, and anti-discrimination rules.
- Small cells, privacy, and category changes in identity and intersectional research.

Research openings

Opening	Publishable contribution
Hidden harms	connect safety, climate, or dignity to labor behavior and welfare
Firm-side measurement	observe tasks, supervisors, evaluations, internal ladders
Identity/treatment mismatch	distinguish legal category, self-identity, perception, and treatment
Search and applications	observe nonapplication, callback, match, and sorting margins
Comparative portability	identify what mechanism travels across institutions
Intersectional gender	use privacy-safe aggregation with honest precision limits
Policy as mechanism test	decompose which margin moves and why

- Reproduce: synthetic application-gap decompositions inspired by Fluchtmann, Glenny, Harmon, and Maibom.
- Diagnose: what application data see and what they miss; compare to matched employer–employee and job-posting designs.
- Transfer: platform within-worker decompositions and posting-policy design logic.
- Optional memo: how evolving identity measurement changes audit or application designs.

- ① State the labor object.
- ② Define the gender concept.
- ③ Name the observed margin.
- ④ Identify the variation.
- ⑤ Diagnose sorting, treatment, and hidden harms.
- ⑥ State the welfare object and limits.
- ⑦ Say what readers should update.

Takeaways

- New data expand the gender-and-labor frontier by making margins visible.
- The research design still turns on identification, measurement, and welfare interpretation.
- The best projects convert empirical shortfalls into disciplined contributions.
- Week 10 uses this toolkit to build student research designs.